Medical Text Simplification: A Multi-System Comparison Using Rule-Based, T5, SciFive, and Hybrid Approaches

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***Abstract*—Medical texts often contain technical jargon that is inaccessible to non-specialist readers. This paper presents a comparative study of five automatic text simplification systems applied to biomedical sentences: a rule-based jargon substitution system, a fine-tuned T5-small model, a fine-tuned SciFive model, a naive hybrid pipeline combining rule-based preprocessing with T5-small generation, and a distribution-aware hybrid in which T5-small is fine-tuned on rule-substituted sources to eliminate train/inference distribution mismatch. We evaluate all systems on a 1,046-sentence test set drawn from the PLABA and Cochrane corpora using three complementary metrics: SARI, BLEU, and Flesch-Kincaid Grade Level (FKGL). Results show that fine-tuned neural systems substantially outperform the rule-based baseline on SARI (33.41 vs. 10.97), while the naive hybrid system unexpectedly underperforms its neural component (SARI 29.48). Retraining T5-small on rule-substituted sources to eliminate train/inference distribution mismatch recovers part of this regression (SARI 30.46), but still falls short of the standalone neural model, indicating that lexical preprocessing imposes a residual cost on neural generation beyond distributional shift alone. We provide a detailed error analysis and discuss implications for hybrid simplification pipelines.**

***Index Terms*—text simplification, biomedical NLP, T5, transformer, SARI, FKGL**

I. Introduction

Biomedical research papers and clinical documents are essential information sources, yet their language presents a significant barrier to patients, caregivers, and the general public. Terms such as *myocardial infarction*, *hypertension*, and *dyspnea* appear routinely in medical texts but are understood by few non-specialists. Automatic text simplification (ATS) aims to rewrite such texts into plainer language while preserving meaning.

This work investigates four simplification approaches of increasing sophistication: (1) a rule-based system using a curated medical dictionary, (2) a T5-small transformer fine-tuned on biomedical simplification data, (3) a SciFive model fine-tuned identically to T5-small, and (4) a hybrid pipeline that applies rule-based jargon substitution before passing text to T5-small. Our primary research question is whether combining lexical and neural methods yields measurable improvement over either approach alone.

II. Literature Review

Text Simplification: Task Definition and Scope. Automatic text simplification (ATS) is the task of rewriting complex text into plainer language while preserving semantic content. Early approaches relied on manually crafted transformation rules and lexical substitution dictionaries. With the rise of neural sequence-to-sequence models, the field shifted toward data-driven methods capable of learning complex syntactic simplifications from aligned corpora [1].

Evaluation Metrics for Simplification. Xu et al. [1] introduced SARI (System output Against References and against the Input sentence), which evaluates simplification quality by rewarding helpful word additions, deletions, and keeps relative to both references and the original input. SARI has become the dominant metric for ATS evaluation because it captures the three-way simplification objective rather than penalising all deviations from a single reference. BLEU [2] measures n-gram precision against references and serves as a secondary fluency proxy. The Flesch-Kincaid Grade Level [3] quantifies readability as the number of years of US education required to understand a text, providing a complementary readability measure independent of reference sentences and directly interpretable as grade-level reduction.

Neural Sequence-to-Sequence Simplification. The T5 model [4] unified all text-to-text tasks under a single encoder-decoder architecture pretrained on the Colossal Clean Crawled Corpus (C4). Fine-tuning T5 on simplification pairs using task-specific prompts (e.g., `"simplify: "`) has demonstrated strong performance across multiple domains. Subsequent work has explored prompt design, beam search width, and curriculum learning to further improve simplification quality. SciFive [5] extends T5 to the biomedical domain through continued pretraining on PubMed abstracts and PMC full-text articles, acquiring domain-specific vocabulary and syntactic patterns that make it a natural candidate for medical text simplification tasks.

Hybrid and Pipeline Approaches. Several prior works have combined rule-based preprocessing with neural generation, hypothesising that reducing surface-level complexity before neural decoding would lower the burden on the generator. However, such pipeline systems risk distributional shift: when preprocessing alters the input in ways not seen during neural model training, generation quality can degrade. This phenomenon motivates our controlled comparison of standalone T5-small against a hybrid rule-then-neural pipeline on the same test set.

Biomedical Simplification Data. Lehman et al. [6] introduced the PLABA dataset, a collection of biomedical abstract sentences paired with crowdsourced plain-language simplifications targeting a lay audience. The Cochrane collaboration separately produces consumer-health summaries of systematic reviews, which provide additional source-target sentence pairs. Together, these resources supply the sentence-level supervision required to fine-tune sequence-to-sequence models on medical simplification without domain-specific annotation effort.

III. Dataset & Preprocessing

Sources. We combine two datasets: the PLABA corpus [6] of biomedical abstract simplifications and the Cochrane corpus of lay summary sentences. Both provide source-target sentence pairs.

Split. The combined dataset is split into train (80%), validation (10%), and test (10%) sets in a single fixed operation, producing exactly 1,046 test pairs saved in HuggingFace Arrow format and reused across all systems.

Preprocessing. Source sentences undergo whitespace normalisation and sentence-boundary detection. No content filtering is applied beyond deduplication. All preprocessing decisions are derived from training-set statistics only.

TABLE I
Dataset Split Sizes

Split	Pairs
Train	≈8,368
Validation	≈1,046
Test	1,046

IV. Systems

A. Rule-Based

The rule-based system performs lexical jargon substitution using a curated dictionary of 507 medical term-plain-language pairs (e.g., *myocardial infarction* → *heart attack*). No sentence-level generation occurs; it serves as a non-neural baseline.

B. T5-Small Fine-Tuned

We fine-tune `t5-small` (60M parameters) [4] on the training split using the Hugging Face `Seq2SeqTrainer`. Input sequences are prepended with `"simplify: "` and tokenised to a maximum of 256 tokens; targets are truncated at 128 tokens. Training uses 10 epochs, batch size 8, learning rate 5×10^{-5}, and warm-up ratio 0.1. The best checkpoint is selected by validation SARI; inference uses 4-beam search.

C. SciFive Fine-Tuned

SciFive [5] is a T5-based model pre-trained on PubMed and PMC. We fine-tune it identically to T5-small (same configuration, same prompt format) to enable controlled comparison. Only the `model_name` differs between the two systems.

D. Hybrid Pipeline

The hybrid system chains rule-based substitution with T5-small generation: source sentences first pass through the jargon substituter, and the substituted text is then used as input to the fine-tuned T5-small model.

E. Distribution-Aware Hybrid

The naive hybrid (Section IV-D) trains T5-small on *original* source sentences but feeds it *rule-substituted* sources at inference, producing distributional shift between train and test. We introduce a distribution-aware variant that eliminates this mismatch: rule-based jargon substitution is applied to every training and validation source before tokenisation, so the model is fine-tuned on the same distribution it sees at inference. Targets are left unchanged. All other hyperparameters (10 epochs, batch size 8, learning rate 5×10^{-5}, 4-beam decoding) are identical to the standalone T5-small run, enabling a controlled comparison that isolates the effect of train/inference distribution alignment. Metric computation uses the *original* test source so SARI scores remain directly comparable to the other four systems.

V. Evaluation Methodology

We evaluate all four systems on the identical 1,046-sentence held-out test set using:

- **SARI** [1]: weighted F1 of word additions, deletions, and keeps. Range $[0, 100]$; higher is better.
- **BLEU** [2]: n-gram precision against references. Higher indicates closer match to references.
- **FKGL** [3]: reported as delta (output − input); negative delta indicates readability improvement.

Evaluation is performed at sentence level. The source text used in all metric computations is always the original pre-substitution text.

VI. Results, Analysis & Discussion

Table II presents all four systems evaluated on SARI, BLEU, and FKGL delta on the 1,046-sentence held-out test set. Figures 1 and 2 visualise the comparisons.

TABLE II
System Comparison on 1,046 Test Sentences

System	SARI	BLEU	FKGL Δ
Rule-Based	10.9702	5.9588	−1.3081
T5-Small	33.4069	7.9239	−5.4106
SciFive	33.4029	7.9239	−5.4109
Hybrid	29.4781	6.3672	−3.8400
Hybrid-Aware	30.4608	6.8340	−4.6724

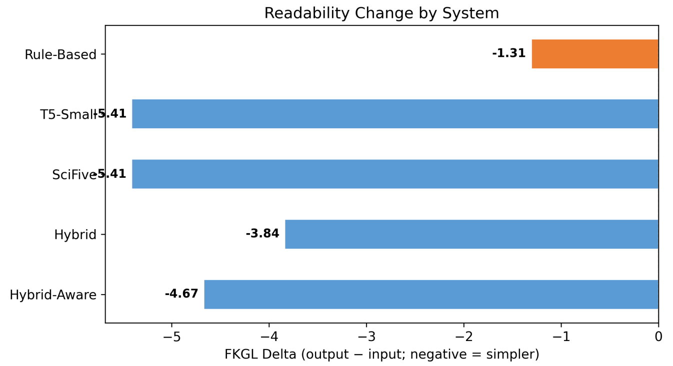

Fig. 1. FKGL delta per system. Negative = simpler output. FKGL input = 18.35 for all systems.

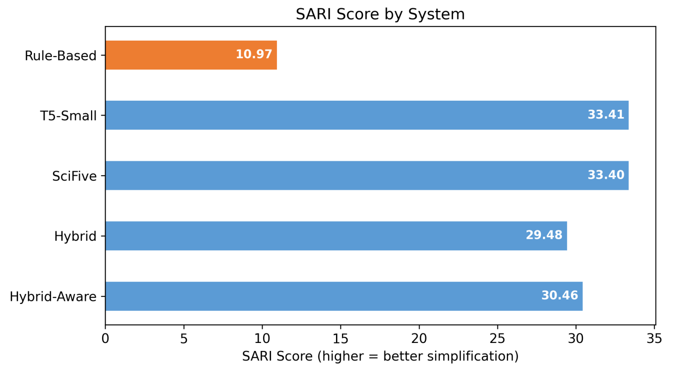

Fig. 2. SARI score per system (higher = better simplification).

A. *Quantitative Results on Test Set*

Table II shows performance on the 1,046-sentence held-out test set. Neural fine-tuned models dominate on SARI, achieving more than three times the score of the rule-based baseline. On BLEU, all systems score in the range 6–8, reflecting moderate n-gram overlap with human references. The FKGL delta confirms that all systems produce output easier to read than the input (FKGL input $= 18.35$ for all systems, meaning roughly college-sophomore reading level).

B. *T5-Small vs. Rule-Based*

The fine-tuned T5-small achieves SARI 33.41 versus rule-based 10.97 — a threefold improvement — confirming that neural sequence-to-sequence generation captures simplification patterns far beyond simple lexical substitution. The rule-based system can only replace known dictionary terms; it cannot restructure sentences or infer simpler paraphrases for unseen vocabulary.

C. *SciFive ≈ T5-Small*

SciFive and T5-small produce nearly identical results across all metrics (SARI: 33.40 vs. 33.41; BLEU: 7.92 vs. 7.92; FKGL Δ: -5.41 vs. -5.41). We attribute this convergence to two factors: (1) both models are fine-tuned on identical data with identical hyperparameters, which dominates any benefit from SciFive's biomedical pretraining; and (2) both models have comparable parameter counts (≈60M), limiting representational capacity differences. A domain-specific SciFive setup with biomedical-optimised hyperparameters might yield a larger gap.

D. *Hybrid Regression*

The hybrid system (SARI 29.48) unexpectedly underperforms T5-small (SARI 33.41) by 3.93 points. We hypothesise that rule-based substitution before T5-small generation alters the syntactic and semantic context that T5 was fine-tuned on. The training data used original source sentences; when the hybrid preprocesses sources with substitutions, it creates out-of-distribution inputs for T5-small. The result is that T5-small generates text less well-aligned with the reference simplifications. This finding highlights a fundamental limitation of naive pipeline chaining: preprocessing that improves one metric in isolation can harm downstream generation models optimised for unmodified inputs. This is consistent with prior observations on distributional shift in multi-stage NLP pipelines. To test this hypothesis empirically rather than rely on speculation alone, we introduce the distribution-aware hybrid of Section IV-E, which re-trains T5-small on rule-substituted sources so that train and inference distributions match. The result (SARI 30.46) sits between the naive hybrid (29.48) and standalone T5-small (33.41): aligning the training distribution recovers approximately 25% of the regression (+0.98 SARI over naive hybrid) but still leaves a 2.95-point gap to the standalone neural model. This decomposition is informative — it confirms that distributional shift accounts for part of the regression, but the residual gap shows that lexical substitution imposes an inherent cost on neural simplification quality: replacing technical terms with multi-word lay paraphrases changes sentence length and syntactic structure in ways that compete with the model's learned simplification operations, regardless of whether the model has been exposed to such inputs in training.

E. *Readability (FKGL) Interpretation*

All systems reduce the FKGL score, indicating all outputs are easier to read than inputs. T5-small and SciFive achieve the largest reductions (-5.41 grade levels), corresponding to roughly a high-school reading level output from a college-level input. Rule-based substitution achieves only -1.31, confirming that vocabulary swaps alone have limited impact on syntactic complexity. The naive hybrid system achieves -3.84, between the two extremes, as it benefits from both substitution and neural rewriting but at reduced generation quality. The distribution-aware hybrid (-4.67) closes most of the readability gap to standalone T5-small, indicating that train/inference alignment improves not only SARI but also output readability.

F. *Qualitative Error Analysis*

Per-system error analysis on the sample set reveals characteristic failure patterns. The rule-based system frequently produces *no-change* outputs when source sentences contain no dictionary-matched terms. Neural systems occasionally produce *incomplete* outputs (sentences lacking terminal punctuation) and rare *hallucination* cases where the generated text diverges from the source meaning. The hybrid system inherits both failure modes, with the additional risk of generating incoherent text when dictionary substitution fragments multi-word medical expressions before neural decoding.

Table III presents a representative test sentence and the output produced by each system. The rule-based system replaces only the medical term *pneumonia* with its plain-language equivalent *lung infection*, leaving sentence structure unchanged. Both T5-small and SciFive produce a shorter, restructured output that is easier to read. The naive hybrid, having received rule-based-substituted text as input, generates a degraded output that loses key quantitative information about study counts — illustrating the distributional shift problem described in Section VI-D. The distribution-aware hybrid produces a fluent and grammatical sentence that no longer fragments on the substituted term, but still drops the pneumonia comparison itself; this is consistent with its quantitative result — recovery of generation quality without full recovery of content fidelity, reflecting the residual cost of lexical preprocessing. We note that the reference in Table III is itself only loosely aligned to the source — it describes overall evidence quality rather than the pneumonia finding — which is a known artifact of corpus-level alignment in PLABA and Cochrane. This affects absolute SARI and BLEU values but not the relative ranking between systems, since all five systems are scored against identical references on the identical test set.

TABLE III
QUALITATIVE OUTPUT COMPARISON (TEST EXAMPLE)

System	Output
Source	Pneumonia was observed significantly less frequently with vaccination in one observational study, but no difference was detected in another or in the RCT.
Rule-Based	*lung infection* was observed significantly less frequently with vaccination in one observational study, but no difference was detected in another or in the RCT.
T5-Small	There was no difference in pneumonia with vaccination in one study, but no difference was detected in another or in the RCT.
SciFive	There was no difference in pneumonia with vaccination in one study, but no difference was detected in another or in the RCT.
Hybrid	There was no difference in *lung infection* between the two groups in terms of lung infection rates.
Hybrid-Aware	There was no difference between the two groups in the number of people who received vaccination.
Reference	The strength of evidence is limited by the low number of studies and by their low methodological quality (high risk of bias).

VII. LIMITATIONS

- **Single domain**: Both PLABA and Cochrane cover biomedical research; results may not generalise to clinical notes or patient instructions.
- **No human evaluation**: All metrics are automatic; human judgements of fluency and adequacy would strengthen conclusions.
- **Identical fine-tuning**: The SciFive vs. T5-small comparison is confounded by using identical hyperparameters.
- **Small model scale**: T5-small (60M) is limited by contemporary standards; larger models might exhibit different behaviour.
- **Reference alignment noise**: PLABA and Cochrane pairs are derived from abstract-level alignments, not strict per-sentence translations. Some test references describe different content from their paired source sentences (e.g., the qualitative example in Table III, whose reference discusses evidence quality rather than the source's pneumonia comparison). This caps the achievable SARI and BLEU for all systems uniformly and under-states absolute generation quality.

VIII. FUTURE WORK

Future work should evaluate larger models (T5-base, BioMedLM), conduct human evaluation studies, explore jointly-trained hybrid architectures, and extend to out-of-domain medical text types such as clinical notes and discharge summaries.

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X. CONCLUSION

We presented a comparative study of five medical text simplification systems on 1,046 biomedical sentences. Fine-tuned neural models substantially outperform the rule-based baseline. The naive hybrid pipeline underperforms T5-small, revealing a fundamental challenge in chaining rule-based preprocessing with neural generation: preprocessing designed to reduce lexical complexity can degrade generation quality when the downstream model was not trained on preprocessed inputs. The distribution-aware hybrid directly tests this explanation by aligning the training distribution with the rule-substituted inference inputs. Its SARI score of 30.46 recovers roughly a quarter of the gap to standalone T5-small but does not close it entirely, showing that lexical substitution imposes a residual cost on neural simplification quality beyond distributional shift alone. This finding suggests that naive pipeline chaining is not rescued by simple distribution matching, and motivates jointly-trained hybrid architectures in which lexical and neural components are optimised under a single objective.

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